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SECTION - A9-1

SUB - SCIENCE FOR ENGINEERING-I

COURSE - B.TECH

CLASS ROLL - 12

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Q-1 Wireless charges work based on Faraday's Law of Electromagnetic induction.

Ans A changing magnetic field created by the charging pad induces an EMF and a corresponding current in the receiver coil of the phone, allowing energy transfer without physical contact.

Q-2 The needle deflects because of Electromagnetic induction.

Ans - moving magnets in and out of a coil changes the magnetic flux linked with the coil. According to Faraday's law, this change in flux induces an electric current, which causes the galvanometer needle to move.

Q-3 It is safe to sit inside a car because of the Faraday cage effect.

Ans - A car has a metal body. When lightning strikes, the electric charges reside on the outer surface of the conductor and do not penetrate the interior. The electric field inside a closed conductor is zero.

Q-4 This is stated by Gauss's law for magnetism.

Ans - This law states that the net magnetic flux through any closed surface is zero ($\oint \mathbf{B} \cdot d\mathbf{A} = 0$). This implies that magnetic monopoles do not exist every north pole is accompanied by a south pole, making the field lines continuous loops.

Q-5 - The law that makes this possible is the law of Reflection?

Ans - Specifically, rear view mirrors are usually convex mirrors. They provide a wide field of view by reflecting light rays from a wide area toward the driver's eyes.

Q-6 A swimming pool looks shallower due to Refraction of light.

Ans - Light rays travelling from the denser medium (water) to the rarer medium (air) bend away from the normal. This causes the bottom of the pool to appear "lifted" to an observer above.

Q-7 This is explained by the zeroth law of thermodynamics.
Ans - The law states that if two systems are in thermal contact, heat will flow from the hotter one to the colder one until they reach thermal equilibrium (the same temperature).

Q-8 Electric heater work based on Joule's law of heating.
Ans When an electric current (I) passes through a conductor with high resistance (R) for a time (t), it produces heat (H). The relationship is given by:

$$H = I^2 R t$$

Q-9 - This is also justified by the zeroth law of thermodynamics.

Ans - The thermometer and the body reach thermal equilibrium. Since the thermometer is calibrated to show its own temperature, and that temperature is now the same as the body's the reading is accurate.

Q-10 Dual nature of Light.

Ans Interference (Young's Double slit) - This phenomena proves it shows that light waves can overlap to create patterns of constructive and destructive interference.

photoelectric effect - This phenomena proves the particle demonstrates that light energy is delivered in discrete "packets" or quanta, which can knock electrons off a metal surface.

Q-11 De Broglie Waves-

Ans Significance - $\lambda = \frac{h}{mv}$

$$K = \frac{1}{2} mv^2$$

divide and multiply by m

$$K = \frac{1}{2} \times (mv)^2 \quad [p = mv]$$

$$\text{So } K = \frac{p^2}{2m}$$

$$p = \sqrt{2mk}$$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mk}} \quad \text{--- (1)}$$

work energy theorem -
 work done = change in K.E.

$$W = K$$

$$W = q \cdot V$$

$$\text{For } e^- \Rightarrow q = e$$

$$K = W = qe$$

$$\text{putting in eqn (1)} \quad \lambda = \frac{h}{\sqrt{2meV}} \approx 12.27$$

$$(a) \text{ For } 50V - \lambda = \frac{12.27}{\sqrt{50}} = 1.735 \text{ \AA}$$

$$(b) 100V - \lambda = \frac{12.27}{\sqrt{100}} = 1.227 \text{ \AA}$$

Q-12

$$\lambda = \frac{h}{\sqrt{2mk}}$$

$$K = 1.25 \text{ keV}$$

$$= 1.25 \times 10^3 \times 1.6 \times 10^{19} \text{ J}$$

$$h = 6.626 \times 10^{-34} \text{ J.s}$$

$$m = 9.11 \times 10^{-31} \text{ k.g}$$

$$\lambda = \frac{6.626 \times 10^{-34}}{\sqrt{2 \times 9.11 \times 10^{-31} \times 1.25 \times 10^3 \times 1.6 \times 10^{19}}}$$

$$\lambda = 3.46 \times 10^{-11} \text{ m}$$

Q-13

$$\lambda = \frac{h}{mv}$$

$$v = 20 \text{ m/s}$$

$$\lambda = \frac{6.626 \times 10^{-34}}{(9.11 \times 10^{-31}) \times 20}$$

$$\lambda = \frac{6.626 \times 10^{-34}}{1.822 \times 10^{-29}}$$

$$\lambda = 3.63 \times 10^{-5} \text{ m}$$

Q-14

Heisenberg uncertainty Principle -

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

Given

$$v = 3 \times 10^4$$

$$\Delta x \geq \frac{h}{4\pi m \Delta v}$$

$$\Delta x = \frac{6.626 \times 10^{-34}}{4 \times 3.14 \times 1.67 \times 10^{-27} \times 3 \times 10^4}$$

$$\Delta x = \frac{6.626 \times 10^{-34}}{6.29 \times 10^{-22}}$$

$$\Delta x \approx 1.05 \times 10^{-12} \text{ m}$$

Q-15 Uncertainty in momentum (Δp) -

$$\Delta p = \frac{h}{2\pi \Delta x} = \frac{6.62 \times 10^{-34}}{2 \times 3.14 \times 10^{-24}} \approx 1.05 \times 10^{-24}$$

Find the actual momentum (p) for a 1 keV electron -

$$p = \sqrt{2mk \cdot E} = \sqrt{2 \times 9.11 \times 10^{-31} \times 1 \times 10^3 \times 1.602 \times 10^{19}}$$

$$= \sqrt{2.918 \times 10^{-46}}$$

$$= 1.708 \times 10^{-23} \text{ k.g m/s}$$

$$\% \text{ uncertainty} = \left(\frac{\Delta p}{p} \right) \times 100$$

$$= \left(\frac{1.05 \times 10^{-24}}{1.708 \times 10^{-23}} \right) \times 100$$

$$\approx 6.1\%$$

Q-16

Given -

$$L = 10^{-8} \text{ cm} = 10^{-10} \text{ m}$$

 $n = 1$ (ground)

 $n = 2$ (first excited)

$$\Delta E = E_2 - E_1 = \frac{(2^2 - 1^2) h^2}{8mL^2}$$

$$\Delta E = \frac{3h^2}{8mL^2} = \frac{3 \times (6.626 \times 10^{-34})^2}{8 \times 9.11 \times 10^{-31} \times (10^{-10})^2}$$

$$\Delta E = 1.808 \times 10^{-17} \text{ J}$$

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$$

$$\Delta E = 112.8 \text{ eV}$$

Q-17

 $n = 1$ (ground)

$$L = 2.5 \times 10^{-10} \text{ m}$$

$$E = \frac{1^2 \times (6.626 \times 10^{-34})^2}{8 \times (9.11 \times 10^{-31}) \times (2.5 \times 10^{-10})^2}$$

$$E_1 = 9.64 \times 10^{-19} \text{ J}$$

$$E_1 \approx 6.02 \text{ eV}$$

Q-18

 $n = 1$ (ground)

$$\Psi_1(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right)$$

$$P = \int_{0.452}^{0.552} |\Psi|^2 dx = \int_{0.452}^{0.552} \frac{2}{L} \sin^2\left(\frac{\pi x}{L}\right) dx$$

$$P = \frac{1}{2} \int_{0.452}^{0.552} \left[1 - \cos\left(\frac{2\pi x}{L}\right)\right] dx \quad \left[\sin^2 \theta = \frac{1 - \cos 2\theta}{2}\right]$$

$$P = \frac{1}{2} \left[x - \frac{L}{2\pi} \sin\left(\frac{2\pi x}{L}\right) \right]_{0.452}^{0.552}$$

$$P = (0.55 - 0.45) - \frac{1}{2\pi} [\sin(1.1\pi) - \sin(0.9\pi)]$$

$$= 0.1 - \frac{1}{2\pi} [-0.309 - 0.309] = 0.1 + \frac{0.618}{2\pi}$$

$$P = 0.198 \text{ or } 19.8\%$$

Q-19 $\Psi = ax$
 Normalization the wave fcn: $\int_0^1 |\Psi|^2 dx = \int_0^1 (ax)^2 dx$

$$= a^2 \left[\frac{x^3}{3} \right]_0^1 = \frac{a^2}{3} = 1$$

$$a = \sqrt{3}$$

$$x = 0.35 \text{ and } x = 0.45$$

$$P = \int_{0.35}^{0.45} 3x^2 dx = \left[x^3 \right]_{0.35}^{0.45}$$

$$P = (0.45)^3 - (0.35)^3 = 0.091125 - 0.042875$$

$$P = 0.04825 \text{ or } 4.825\%$$